PF3 Worked Examples: Inverse Trigonometric Functions

Recording companion for the PF3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Which angle has this sine, cosine, or tangent, and is that angle inside the function's range? Find an angle on the unit circle with the right value. Then check the range: arcsine and arctangent give angles from negative pi over 2 to pi over 2, the right half of the circle. Arccosine gives angles from 0 to pi, the top half. If your angle is outside, find the one inside that has the same value.

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Example 1: Evaluate 1

$$\arccos(-1)$$

Question

What is the exact value?

A. $-\pi$
B. 0
C. $\pi/2$
D. π

Answer and reasoning

Cosine is the x-coordinate on the unit circle. It is -1 at the far left point, angle π. Arccosine gives angles from 0 to π, and π is the end of that range, so it counts. $-\pi$ points to the same spot, but arccosine is never negative; 0 is the far right point, where the x-coordinate is 1, not -1; and $\pi/2$ is the top of the circle, where the x-coordinate is not -1 either. **The value is π.**

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Example 2: Evaluate 2

$$\arccos\left(-\frac{\sqrt{2}}{2}\right)$$

Question

What is the exact value?

- **A.** $-\pi/4$
- **B.** $3\pi/4$
- **C.** $5\pi/4$
- **D.** $\pi/4$

Answer and reasoning

The input is $-(\sqrt{2})/2$. Arccosine gives angles from 0 to π, the top half of the circle. Cosine is negative on the left side, so the angle is in quadrant II: $3\pi/4$. The angle $5\pi/4$ also has that cosine but is on the bottom half. Arccosine is never negative, which rules out $-\pi/4$, and cosine at $\pi/4$ is positive. **The value is $3\pi/4$.**

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Example 3: Evaluate 3

$$\arctan(-\sqrt{3})$$

Question

What is the exact value?

A. $2\pi/3$
B. $-\pi/6$
C. $5\pi/3$
D. $-\pi/3$

Answer and reasoning

Tangent is $\sqrt{3}$ at $\pi/3$. Arctangent gives angles between $-\pi/2$ and $\pi/2$, and a negative input gives a negative angle, so the answer is $-\pi/3$. The angles $2\pi/3$ and $5\pi/3$ have the same tangent but are outside the range. Tangent at $-\pi/6$ is $-(\sqrt{3})/3$. **The value is $-\pi/3$.**

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Example 4: Evaluate 4

$$\arcsin(-1/2)$$

Question

What is the exact value?

A. $-\pi/6$
B. $7\pi/6$
C. $11\pi/6$
D. $-\pi/3$

Answer and reasoning

Sine is $1/2$ at $\pi/6$. A negative input to arcsine gives a negative angle, in quadrant IV: $-\pi/6$. The angle $11\pi/6$ points the same way, but arcsine's range only goes down to $-\pi/2$, so the answer is written as $-\pi/6$. Sine at $-\pi/3$ is $-(\sqrt{3})/2$. The angle $7\pi/6$ also has sine $-1/2$, but it is on the left half of the circle. **The value is $-\pi/6$.**

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Example 5: Value 1

$$\arccos(-1/2)$$

Question

What is the exact value?

A. $-\pi/3$
B. $4\pi/3$
C. $2\pi/3$
D. $\pi/3$

Answer and reasoning

Cosine is $1/2$ at $\pi/3$. A negative cosine means the left side of the circle, and arccosine stays on the top half, so the angle is in quadrant II: $\pi-\pi/3=2\pi/3$. The answer $-\pi/3$ is below the x-axis, but arccosine never leaves the top half. The angle $4\pi/3$ is on the bottom half, and cosine at $\pi/3$ is positive. **The value is $2\pi/3$.**

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Example 6: Value 2

$$\arctan((\sqrt{3})/3)$$

Question

What is the exact value?

A. $\pi/3$
B. $\pi/6$
C. $7\pi/6$
D. $\pi/4$

Answer and reasoning

The input is $(\sqrt{3})/3$, which is the same as $1/\sqrt{3}$. At $\pi/6$, sine is $1/2$ and cosine is $(\sqrt{3})/2$, so tangent is $1/\sqrt{3}$. At $\pi/3$ tangent is $\sqrt{3}$ instead, and at $\pi/4$ it is 1. The angle $7\pi/6$ has the same tangent but is outside the range. **The value is $\pi/6$.**

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Example 7: Value 3

$$\arccos(0)$$

Question

What is the exact value?

A. 0
B. $3\pi/2$
C. $-\pi/2$
D. $\pi/2$

Answer and reasoning

Cosine is the x-coordinate on the unit circle, and it is 0 at the top and bottom of the circle. Arccosine only uses the top half, from 0 to π, so the answer is the top: $\pi/2$. The answer 0 is arcsine of 0, not arccosine. The angles $3\pi/2$ and $-\pi/2$ point to the bottom, outside the range. **The value is $\pi/2$.**

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Example 8: Value 4

$$\arctan(-1)$$

Question

What is the exact value?

A. $3\pi/4$
B. $-5\pi/4$
C. $-\pi/4$
D. $\pi/4$

Answer and reasoning

Tangent is 1 at $\pi/4$, where sine and cosine are equal. A negative input to arctangent gives a negative angle, between $-\pi/2$ and 0: $-\pi/4$. The angles $3\pi/4$ and $-5\pi/4$ also have tangent -1, but they are outside arctangent's range; and $\pi/4$ drops the sign. **The value is $-\pi/4$.**

Example 9: Range 1

$$f(x)=\arcsin(x)$$

Question

What is the range of this function?

A. $[0,\pi/2]$
B. $[-\pi/2,\pi/2]$
C. $(-\pi/2,\pi/2)$
D. $[-\pi,\pi]$

Answer and reasoning

Arcsine gives angles on the right half of the unit circle, from $-\pi/2$ to $\pi/2$, and it reaches both ends: $\arcsin(1)=\pi/2$ and $\arcsin(-1)=-\pi/2$. So the brackets are square, not round. $[0,\pi/2]$ forgets that negative inputs give negative angles, and $[-\pi,\pi]$ is a whole turn, but arcsine gives just one angle for each input. **The range is $[-\pi/2,\pi/2]$.**

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Example 10: Domain 1

$$g(x)=\arccos(x)$$

Question

What is the domain of this function?

- **A.** $[-1,1]$
- **B.** $[0,1]$
- **C.** $(-\infty,\infty)$
- **D.** $[-\pi/2,\pi/2]$

Answer and reasoning

The input to arccosine is a cosine value, and cosine is always between -1 and 1, negative on the left half of the circle. So exactly those inputs make sense: $\arccos(2)$ has no answer. $[0,1]$ leaves out the negative cosines; $(-\infty,\infty)$ would need cosine to take every value; and $[-\pi/2,\pi/2]$ is a set of angles, but the input here is a cosine value, a number. **The domain is $[-1,1]$.**

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Example 11: Range 2

$$h(x)=\arctan(x)$$

Question

What is the range of this function?

A. $[-\pi/2,\pi/2]$
B. $(-\infty,\infty)$
C. $(-\pi/2,\pi/2)$
D. $(0,\pi)$

Answer and reasoning

Arctangent gives angles on the right half of the circle. But tangent is undefined at $\pm\pi/2$, so arctangent never reaches those angles; its graph only gets closer and closer to them. The ends are left out, so the brackets are round, not square. $(-\infty,\infty)$ is what goes in, the domain, not what comes out; and $(0,\pi)$ is the top half of the circle, but a negative input to arctangent gives a negative angle. **The range is $(-\pi/2,\pi/2)$.**

Example 12: Range 3

$$g(x)=\arccos(x)$$

Question

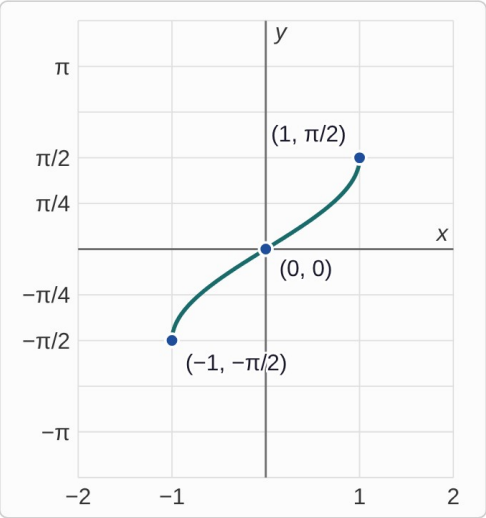
What is the range of this function?

A. $[-\pi/2,\pi/2]$
B. $[0,\pi]$
C. $[0,2\pi)$
D. $(0,\pi)$

Answer and reasoning

Arccosine gives angles on the top half of the circle, from 0 to π, and it reaches both ends: $\arccos(1)=0$ and $\arccos(-1)=\pi$. So the brackets are square. $[-\pi/2,\pi/2]$ is the right half, which would give negative angles, and arccosine is never negative; $[0,2\pi)$ is the whole circle, but each input gets just one angle; and $(0,\pi)$ leaves out the ends, which are reached. **The range is $[0,\pi]$.**

Example 13: Graph 1



Plotted data: a curve that exists only from x = −1 to x = 1, with solid dots at both ends: (−1, −π/2) and (1, π/2). It passes through (0, 0) and rises from left to right. It is steepest near its two ends and flattest in the middle. Nothing is drawn to the left of x = −1 or to the right of x = 1.

Question

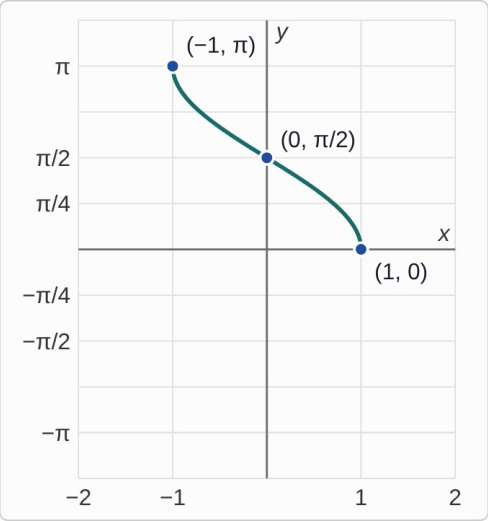
Which function has this graph?

A. $y=\arccos(x)$
B. $y=\arctan(x)$
C. $y=\sin(x)$
D. $y=\arcsin(x)$

Answer and reasoning

The curve only exists for inputs from -1 to 1, so it is arcsine or arccosine. It rises and passes through $(0, 0)$, and $\arcsin(0)=0$. The outputs run from $-\pi/2$ to $\pi/2$, the range of arcsine. Arccosine is never negative, but this curve goes below the x-axis; arctangent would accept every input; and sine would repeat forever left and right. **The graph is $y=\arcsin(x)$.**

Example 14: Graph 2



Plotted data: a curve that exists only from x = −1 to x = 1, with solid dots at both ends: (−1, π) and (1, 0). It passes through (0, π/2) and falls from left to right. It is steepest near its two ends. Nothing is drawn to the left of x = −1 or to the right of x = 1.

Question

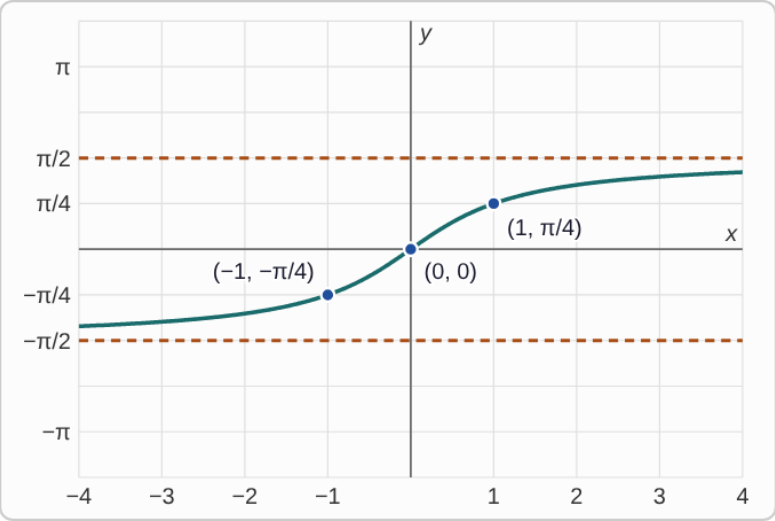
Which function has this graph?

- **A.** $y=\arccos(x)$
- **B.** $y=\arcsin(x)$
- **C.** $y=\arctan(x)$
- **D.** $y=\arcsin(x)+\pi/2$

Answer and reasoning

The inputs run from -1 to 1, and every output is between 0 and π, never negative. That is the range of arccosine. Check the points: $\arccos(1)=0$, $\arccos(0)=\pi/2$, and $\arccos(-1)=\pi$. Arcsine and arctangent both give negative outputs for negative inputs, and arctangent accepts every input. $\arcsin(x)+\pi/2$ also has outputs from 0 to π, but it rises from left to right, and this curve falls. **The graph is $y=\arccos(x)$.**

Example 15: Graph 3



Plotted data: dashed horizontal lines at y = −π/2 and y = π/2. A curve runs across the whole grid, from x = −4 to x = 4, rising from left to right. It passes through (−1, −π/4), (0, 0), and (1, π/4). Toward the left it levels off, approaching the lower dashed line, and toward the right it approaches the upper one, without touching either.

Question

Which function has this graph?

A. $y=\arcsin(x)$
B. $y=\sin(x)$
C. $y=\arctan(x)$
D. $y=\arccos(x)$

Answer and reasoning

The curve accepts every input, so it cannot be arcsine or arccosine, which stop at -1 and 1. It levels off toward $y=\pm\pi/2$ without reaching them, which is the range of arctangent. Check: $\arctan(1)=\pi/4$. Sine would wave up and down between -1 and 1 forever. **The graph is $y=\arctan(x)$.**

Example 16: Slip 1

$$\arcsin((\sqrt{3})/2)=2\pi/3$$

Question

A student wrote this because $\sin(2\pi/3)=(\sqrt{3})/2$. What is the mistake, if any?

A. There is no mistake.
B. The sine of that angle is really one half, so the angle is wrong.
C. The other angle with this sine works instead: $4\pi/3$.
D. Arcsine stays on the right half, so the answer is $\pi/3$.

Answer and reasoning

The student's fact is right: $\sin(2\pi/3)=(\sqrt{3})/2$, not one half. But $2\pi/3$ is outside arcsine's range. Arcsine picks the one angle on the right half of the circle with that sine. The sine of $4\pi/3$ is negative. **The correct value is $\pi/3$.**

Example 17: Slip 2

$$\arccos(-(\sqrt{2})/2) = -\pi/4$$

Question

A student wrote this. What is the mistake, if any?

A. There is no mistake.
B. Cosine is negative there, so the answer is $5\pi/4$.
C. Arccosine is never negative; the answer is $3\pi/4$.
D. Arccosine ignores the sign, so the answer is $\pi/4$.

Answer and reasoning

Arccosine gives angles from 0 to π, the top half of the circle, so it is never negative. A negative cosine puts the angle on the left side of the top half: $3\pi/4$. The student used arcsine's rule: $-\pi/4$ is outside arccosine's range, and its cosine is positive, so it does not even match the negative input. The angle $5\pi/4$ is on the bottom half, and ignoring the sign gives $\pi/4$, whose cosine is positive. **The correct value is $3\pi/4$.**

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Example 18: Slip 3

$$\arcsin(2)$$

Question

A student says this has no value. What is the mistake, if any?

A. There is no mistake.
B. Since 2 is close to 1, it equals $\pi/2$.
C. It equals sin(2), which is about 0.91.
D. It equals one full turn around the circle, 2π.

Answer and reasoning

Sine is never more than 1, so no angle has a sine of 2. The domain of arcsine is $[-1,1]$, and 2 is outside it. The student is right. $\pi/2$ has sine exactly 1, and close is not enough; $\sin(2)$ goes the wrong way, from an angle to a value; and a full turn, 2π, has sine 0.
There is no mistake: arcsine of 2 is undefined.

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Example 19: Slip 4

$$h(x)=\arctan(x)$$

Question

A student says the range of this function is $[-\pi/2,\pi/2]$, with square brackets. What is the mistake, if any?

A. There is no mistake.
B. The range should be all real numbers, like the domain.
C. The range should match arccosine, $[0,\pi]$.
D. The ends are never reached, so use round brackets.

Answer and reasoning

Tangent is undefined at $\pm\pi/2$, so arctangent never outputs those angles. Its graph only levels off toward them. Leaving the ends out takes round brackets. All real numbers is the domain, not the range; and $[0,\pi]$, the top half, would leave out the negative angles that negative inputs give. **The range is $(-\pi/2,\pi/2)$.**

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Example 20: Slip 5

$$\sin^{-1}(x) = 1/\sin(x)$$

Question

A student wrote this. What is the mistake, if any?

A. There is no mistake.
B. The right side should be 1 over the cosine of x instead.
C. The negative 1 means arcsine here, not one over sine.
D. The right side should be the negative of the sine of x.

Answer and reasoning

On a trig function, the -1 is not an exponent: it names the inverse function, arcsine, which takes a value and gives back an angle. One over sine is cosecant, a different function. For example, arcsine of 1 is $\pi/2$, but one over sine of 1 is about 1.19. One over cosine is secant, and the negative of sine, $-\sin(x)$, is a reflection of the sine graph; neither is what the -1 means. **The notation means arcsine, not one over sine.**

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